

Brain Tumor Classification

DATASCI 281: Foundations of Computer Vision

Aaron Luong, Amanda Chung, Caitlin Gainey, Clara Rhoades

Introduction

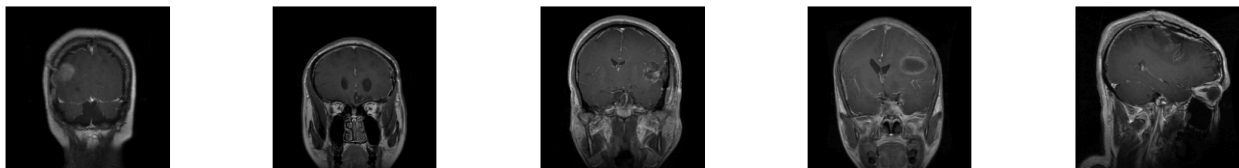
Brain tumors affect approximately 90,000 Americans annually and over 308,000 people worldwide each year (Cohen-Gadol). Malignant tumors have a five-year survival rate of around 36%, although this figure varies substantially by tumor type and stage at diagnosis. Accurate and timely classification of tumor type from MRI imaging is a critical step in treatment planning, yet tumor classification remains challenging due to high variability in tumor morphology, imaging conditions, and anatomical context across patients.

This project applies classical computer vision techniques to the problem of brain tumor classification from MRI scans. We target four clinically relevant categories: glioma, meningioma, pituitary tumor, and no tumor. Our key contribution is a controlled study of feature representation in brain tumor classification, evaluating how gradient-based (HOG), texture-based (LBP), and deep-learned (ResNet) features capture complementary aspects of MRI data, and how their combination influences downstream classification performance. We tune feature and hyperparameter configurations for both logistic regression and SVM, optimizing for validation accuracy, and then analyze the efficiency-accuracy tradeoff of the top-performing models.

Dataset

We leveraged the Brain Tumor MRI Dataset from Kaggle, which contains 7,200 human brain MRI images across four classes: no tumor, glioma, meningioma, and pituitary tumor. Prior to pre-processing, our raw dataset was well-balanced, with 1,800 images per class.

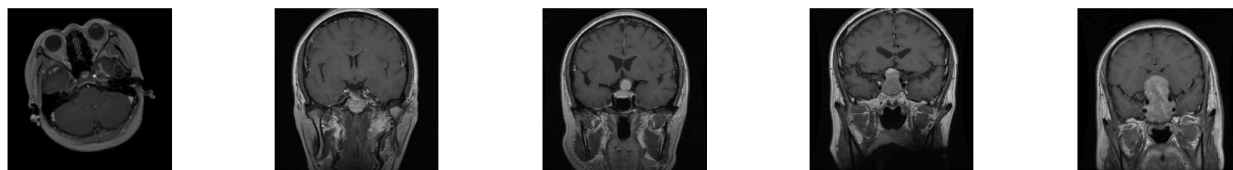
Sample MRI Images — Glioma



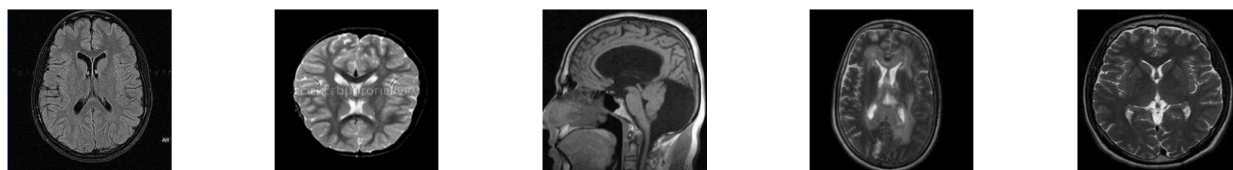
Sample MRI Images — Meningioma



Sample MRI Images — Pituitary



Sample MRI Images — No Tumor



Images vary considerably in resolution, from roughly 150×198 to $1,375 \times 1,446$ pixels, and are captured across axial, coronal, and sagittal planes. Tumors differ in location, size, shape, and internal texture across classes, and normal anatomical variation is present throughout the dataset.

Exploratory Data Analysis & Pre-Processing

Before any preprocessing, we performed a series of data quality checks on the full dataset, including duplicate detection, resolution, pixel quality and contrast, and color channel analysis. Then, we implemented a pre-processing pipeline to ensure consistent size, resolution, and number of channels across images. Finally, as part of our exploratory analysis, we implemented a convolutional neural network without any feature engineering to give us an idea of how much discriminative signal was learnable directly from the pre-processed training data. This served as a data-driven baseline that helped identify which class distinctions could be easier or more difficult and where feature engineering could be most impactful.

Duplicate Detection

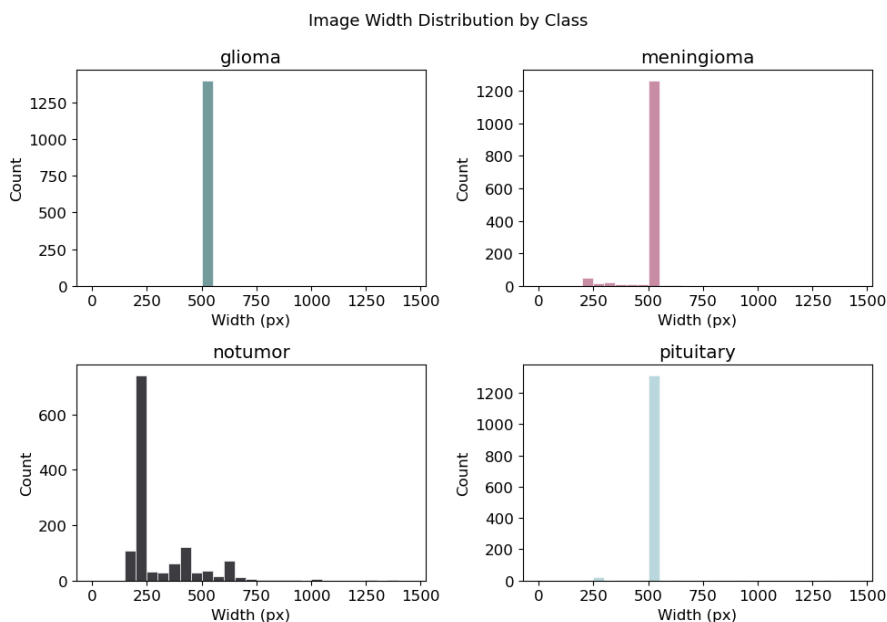
An MD5 hash comparison across all 7,200 images identified 187 duplicates: 171 within the training set, 16 within the test set, and 0 across the train-test boundary. The absence of cross-split duplicates indicates that the original Kaggle partitions were free of train-test contamination. Interestingly, duplicates were heavily concentrated in the no-tumor class (119 of the 187 removals), perhaps due to no tumor images being sourced differently than the other three classes.

Resolution, Pixel Quality, and Contrast

Image resolutions range widely from 150×198 to $1,375 \times 1,446$ pixels, with a median of 512×512 and a median aspect ratio of 1.0. The order-of-magnitude spread means we can't feed raw images to a classifier without resizing them. The median 512 and aspect ratio 1.0 tell us the

direction of the resize: most images are square and larger than our target, so most images are downsampled.

Per-class resolution distributions reveal that the low-resolution images are concentrated almost entirely in the no-tumor class. Interestingly, both the smallest image (150×198) and the largest image (1,375×1,446) come from the no-tumor class, supporting the theory that the no-tumor images were drawn from a less uniform set of sources than the other three classes. This convergence matters because resolution itself becomes a confounding feature: without resizing, a classifier could exploit input dimensions as a spurious correlation of class label rather than learning the underlying anatomical distinctions. Resizing every image to a fixed 128×128 target removes this confound and forces the model to rely on image content.



All images are stored as uint8 with pixel values spanning the full 0–255 range, indicating consistent encoding across sources. A per-image dynamic-range check found no images using less than 30% of that range, and a full-dataset scan for near-blank or low-contrast scans (mean intensity outside [15, 240] or standard deviation below 10) flagged a single suspect image. Given that this was not a widespread problem, we did not exclude the single image from the dataset. Together these checks indicated that contrast was already adequate across the dataset, and we therefore did not apply global contrast normalization (e.g., CLAHE) in the preprocessing pipeline.

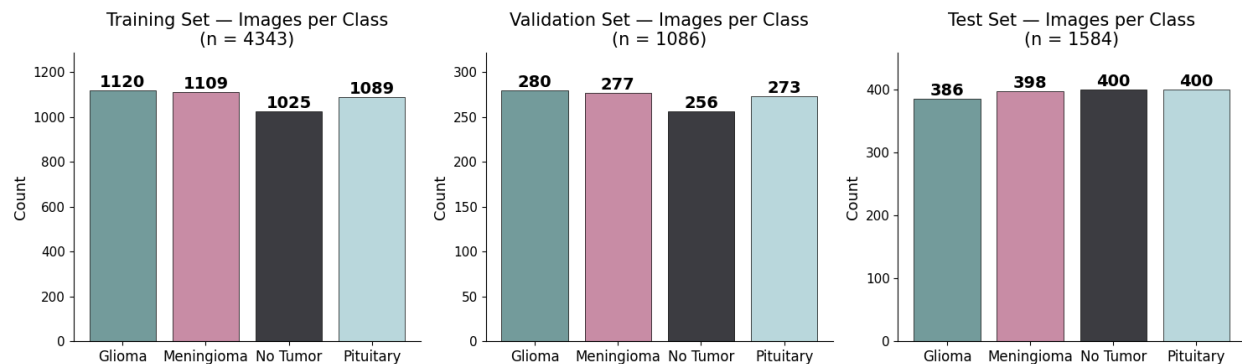
Color Channel Analysis

All images are stored as 3-channel files despite MRI being an inherently grayscale modality. A full-dataset channel analysis found that 7,071 images have identical channels, while 129 images exhibit genuine inter-channel differences. All 129 belong to the no-tumor class. Of these, 113 were classified as real color content rather than JPEG compression artifacts (max inter-channel difference > 5). Because this color signal is entirely class-specific, retaining RGB inputs would

let a classifier exploit color as a confounding feature rather than learning anatomical structure. We therefore load all images as single-channel grayscale during preprocessing.

Pre-Processing Pipeline

Following deduplication, each image was loaded as single-channel grayscale via PIL, resized to 128×128 using Lanczos interpolation, and normalized to the [0, 1] range as a float32 array. The 128×128 target resolution preserves the spatial detail required by our hand-engineered features (HOG and LBP) while keeping feature dimensionality and extraction time tractable; ResNet-18 inputs are subsequently resized to the network's native 224×224 within its own transform pipeline. Lanczos interpolation was chosen over simpler bilinear or area methods to better preserve edge structure during downsampling, which matters for the gradient-based HOG descriptor in particular.



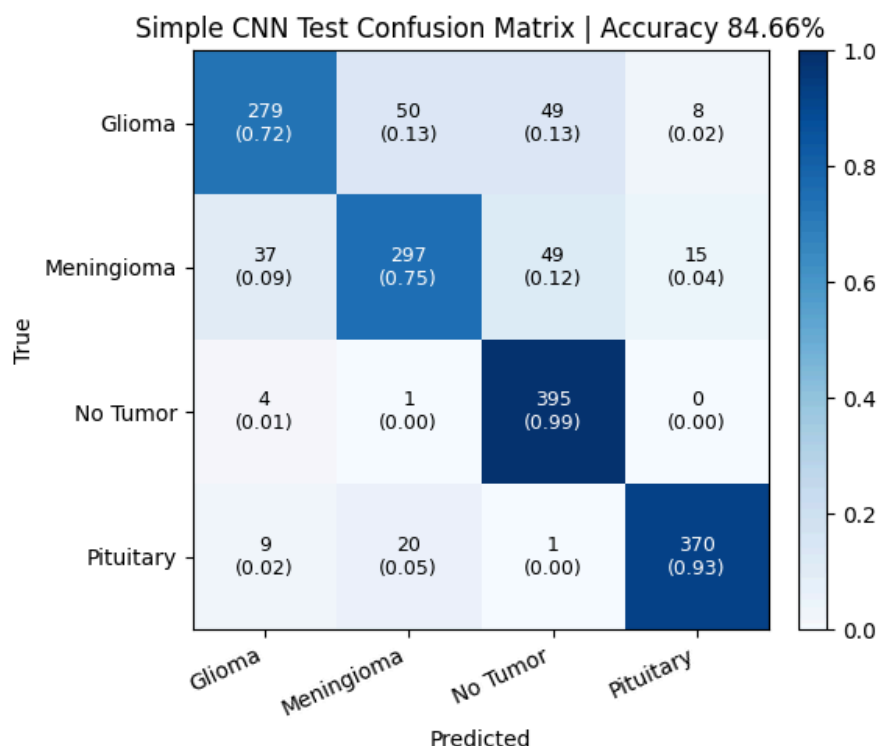
The deduplicated training pool consists of 5,429 images. We then divide it into train and validation sets using a stratified 80/20 split, yielding final counts of 4,343 training, 1,086 validation, and 1,584 test images. Per-class training counts are 1,120 (glioma), 1,109 (meningioma), 1,025 (no tumor), and 1,089 (pituitary).

Convolutional Neural Network

To probe how much discriminative signal is learnable directly from preprocessed pixel intensities before any feature engineering, we trained a small CNN end-to-end on the same grayscale 128×128 inputs and the same train/validation/test partitioning used by our main pipeline. The model is a minimal three-block convolutional architecture (Conv–ReLU–MaxPool with 16, 32, and 64 channels) followed by a small fully-connected head with dropout, totaling roughly 1.1M parameters. This means it was large enough to learn meaningfully from the data but still an order of magnitude smaller than the pretrained ResNet-18 used downstream. Training used Adam ($\text{lr}=1\text{e-}3$), batch size 64, cross-entropy loss, and early stopping on validation loss with patience of four epochs; training converged in 14 epochs and ~11 seconds on an A100 GPU.

The CNN reached 84.66% test accuracy, with strong performance on no-tumor (98.8% recall) and pituitary (92.5% recall) and weaker performance on glioma (72.3% recall) and meningioma

(74.6% recall). This pattern—easier separation of structurally distinct classes, and persistent confusion between glioma and meningioma—pointed to an intrinsic visual overlap between these two tumor types, and shaped our expectations for which class pairs the downstream classical pipeline would likely struggle to distinguish. The CNN result therefore serves as a useful floor: it indicates the level of accuracy obtainable from end-to-end pixel-level learning with minimal architectural commitment, and motivates the use of more expressive representations such as deep ResNet features and complementary hand-engineered descriptors in the classical pipeline to push beyond it.



While end-to-end deep learning models are a natural approach for this task, our goal is to isolate the impact of feature representation under a controlled modeling framework. Using fixed feature extractors allows for a more direct comparison between hand-engineered and learned representations.

Tumor Detection Experiments

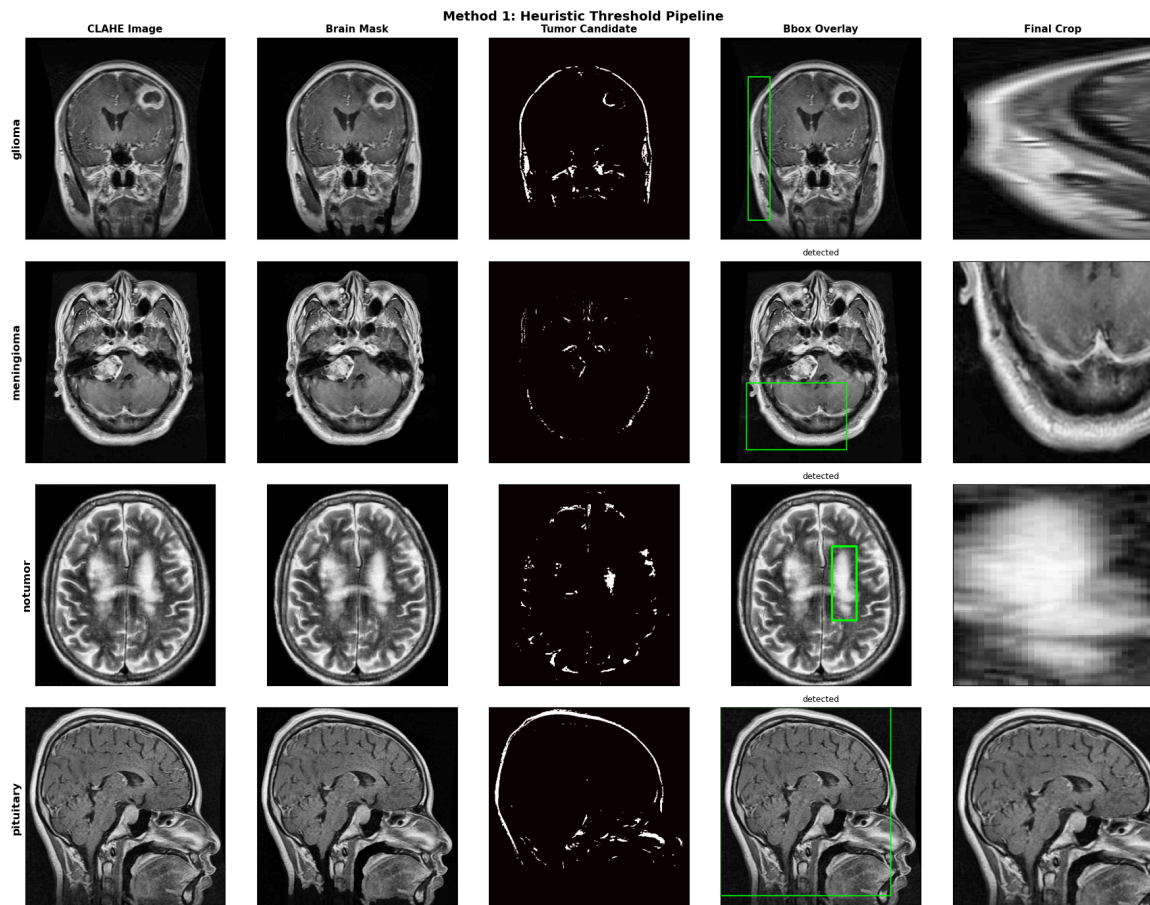
Before settling on whole-image classification, we explored whether automated tumor localization could be used as a preprocessing step to crop regions of interest and reduce background noise in the feature vectors. We explored five heuristic approaches, including pretrained segmentation models, intensity thresholding, bilateral symmetry differencing, skull stripping, and combinations of these approaches.

Pretrained Segmentation (U-Net / MONAI)

We investigated pretrained BraTS segmentation models but found them incompatible with our dataset: they expect 3D multi-modal MRI (T1, T1ce, T2, FLAIR), whereas our dataset consists of 2D single-channel slices. Training a segmentation model from scratch was not feasible without pixel-level annotations.

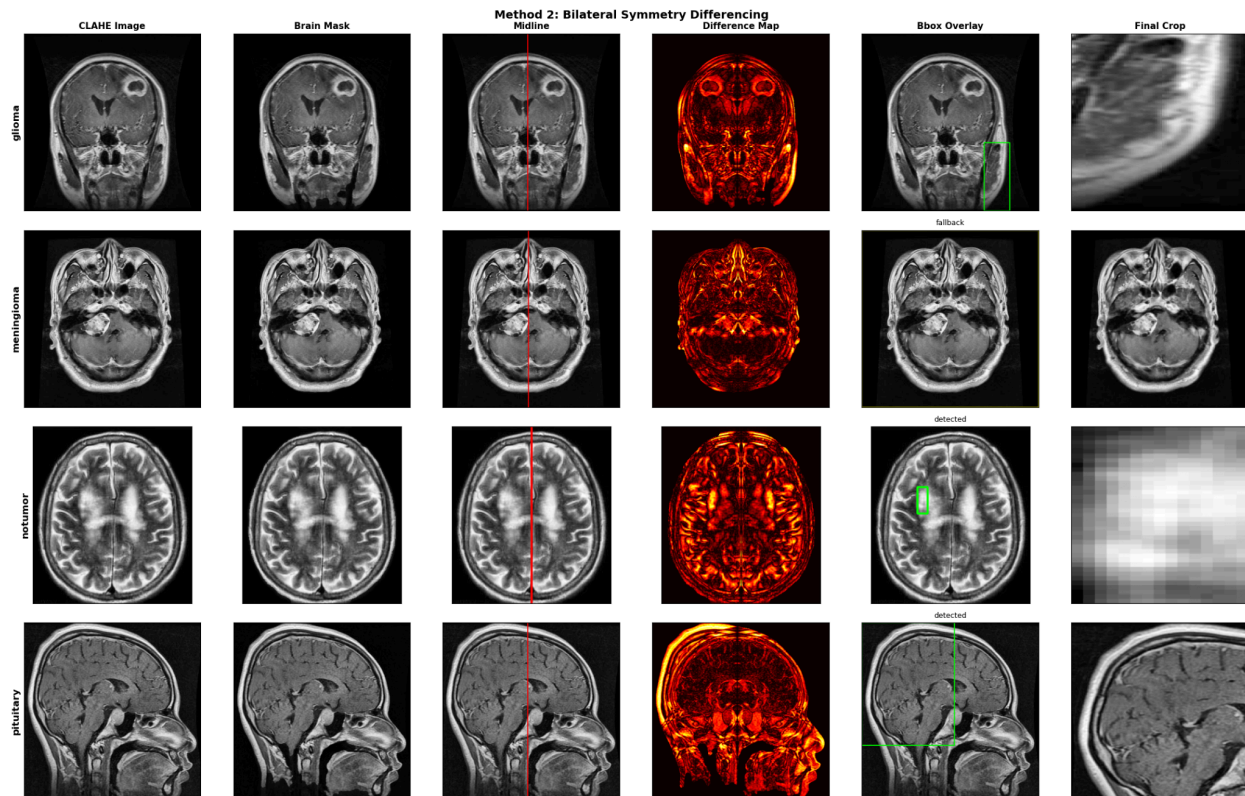
Intensity Thresholding

We applied Otsu-based skull stripping followed by thresholding bright pixels within the brain mask ($\text{mean} + k \times \text{std}$) and selected the largest connected component as the tumor candidate. Candidate maps consistently activated on ventricles, white matter, and cortical surfaces alongside any tumor signal. Glioma crops were narrow, unusable slivers; pituitary sagittal views produced near-full-image bounding boxes due to skull leakage. Raising k from 1.2 to 2.0 had minimal effect. A second iteration used raw grayscale for thresholding, an eroded brain mask to exclude skull edges, and a maximum region area cap of 35%. Pituitary crops improved, but glioma and meningioma failures persisted.



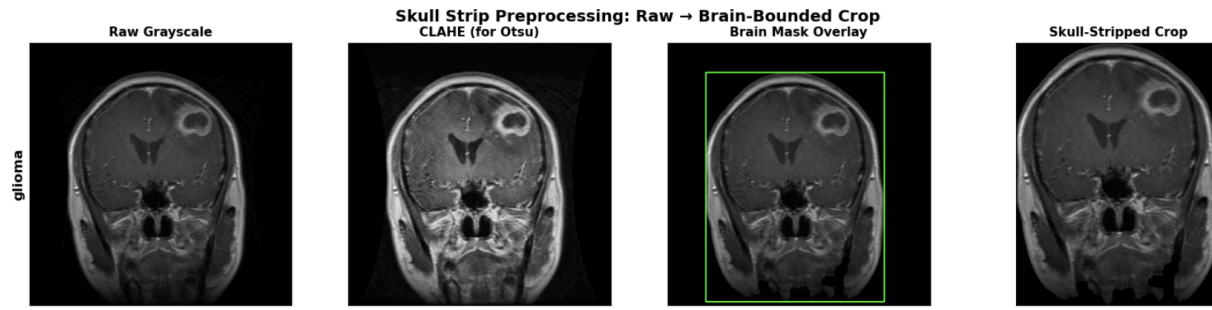
Bilateral Symmetry Differencing

We estimated the brain midline, flipped the image horizontally, and thresholded the absolute difference map to identify asymmetric regions. CLAHE pre-processing made difference maps uniformly active because local contrast enhancement amplified normal tissue asymmetry, making it indistinguishable from tumor signal. Sagittal pituitary views have no bilateral symmetry to exploit, making this approach inapplicable to that class. A second iteration used raw grayscale differencing with Gaussian blur on the difference map, which reduced noise but spread diffuse natural asymmetry into false detections. Meningioma detection improved marginally but remained imprecise.



Skull Stripping

We applied Otsu thresholding with morphological operations to isolate brain tissue. Otsu separates dark background from bright foreground but cannot distinguish skull bone from brain tissue, so the output removed black padding but left the skull intact. True skull stripping requires tools with shape models and atlas priors (e.g., FSL BET, FreeSurfer), which are incompatible with our 2D single-channel pipeline.



Findings: Tumor Detection Experiments

The fundamental limitation across all heuristic approaches is the absence of segmentation ground truth. Without annotations, there is no supervision signal to distinguish tumor tissue from normal bright structures, and the "largest blob" selection heuristic is frequently wrong. We discussed this exploration with the course instructor, who confirmed that pivoting to whole-image classification was appropriate given the dataset constraints. Our final pipeline classifies on full preprocessed images.

Feature Extraction

Feature Selection and Motivation

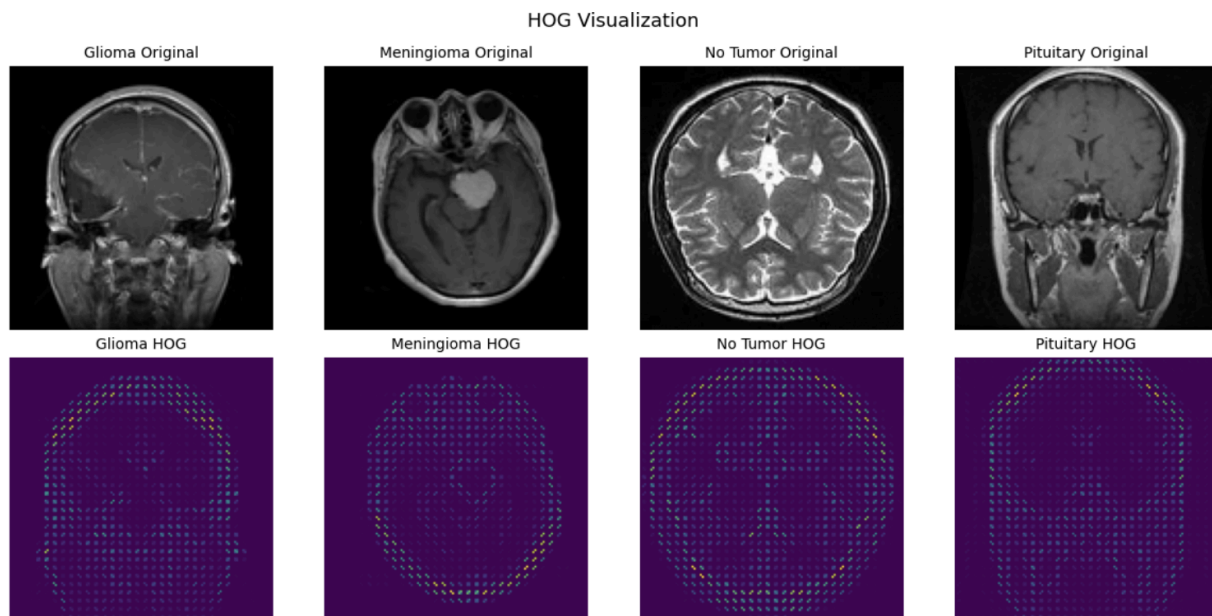
Selecting appropriate features requires understanding which classes are visually distinguishable and which are not. The figure below shows the per-class mean training image after preprocessing. No-tumor and pituitary scans average to structurally distinct images: no-tumor preserves visible sulci and midline anatomy, while pituitary scans show a compact, inferiorly-centered region of signal. Glioma and meningioma, by contrast, produce nearly identical diffuse averages with no clear structural signature. This points to a fundamental asymmetry in the problem—global shape and boundary cues should suffice to separate the structurally distinct class pairs, but the harder pair (glioma vs. meningioma) likely requires finer-grained features that capture local texture or higher-level semantic content. The CNN baseline corroborates this point already: it separates no-tumor and pituitary cleanly while struggling on glioma and meningioma.



We therefore selected three complementary feature representations: a gradient-based descriptor (HOG), a texture descriptor (LBP), and a deep semantic descriptor (ResNet-18). Each targets a different regime of the problem, and we concatenate them into combined feature vectors, HOG+LBP and HOG+LBP+ResNet-18 (also referred to as AllFeat), so that downstream classifiers can exploit the union. When holding all other elements of the modeling pipeline constant, we isolate the effect of the feature choice, enabling a direct comparison between hand-engineered features (HOG and LBP) and learned representations (ResNet18), as well as their combinations.

Histogram of Oriented Gradients (HOG)

HOG encodes local gradient orientations into per-cell histograms, capturing the edge and boundary structure of an image at a chosen spatial granularity. We extract HOG with 12 orientation bins, 4×4-pixel cells, 2×2-cell blocks, and L2-Hys block normalization, yielding a 46,128-dimensional feature vector per image. The HOG visualization confirms that the descriptor activates strongly along skull boundaries and major intracranial structures, with some class-specific patterns visible in the meningioma example, shown as more localized activation near the tumor region. In contrast, the pituitary example shows more diffuse activations and lacks a clearly localized pattern, highlighting that HOG excels at capturing global edge structure and has limitations in isolating smaller or less distinct tumors.



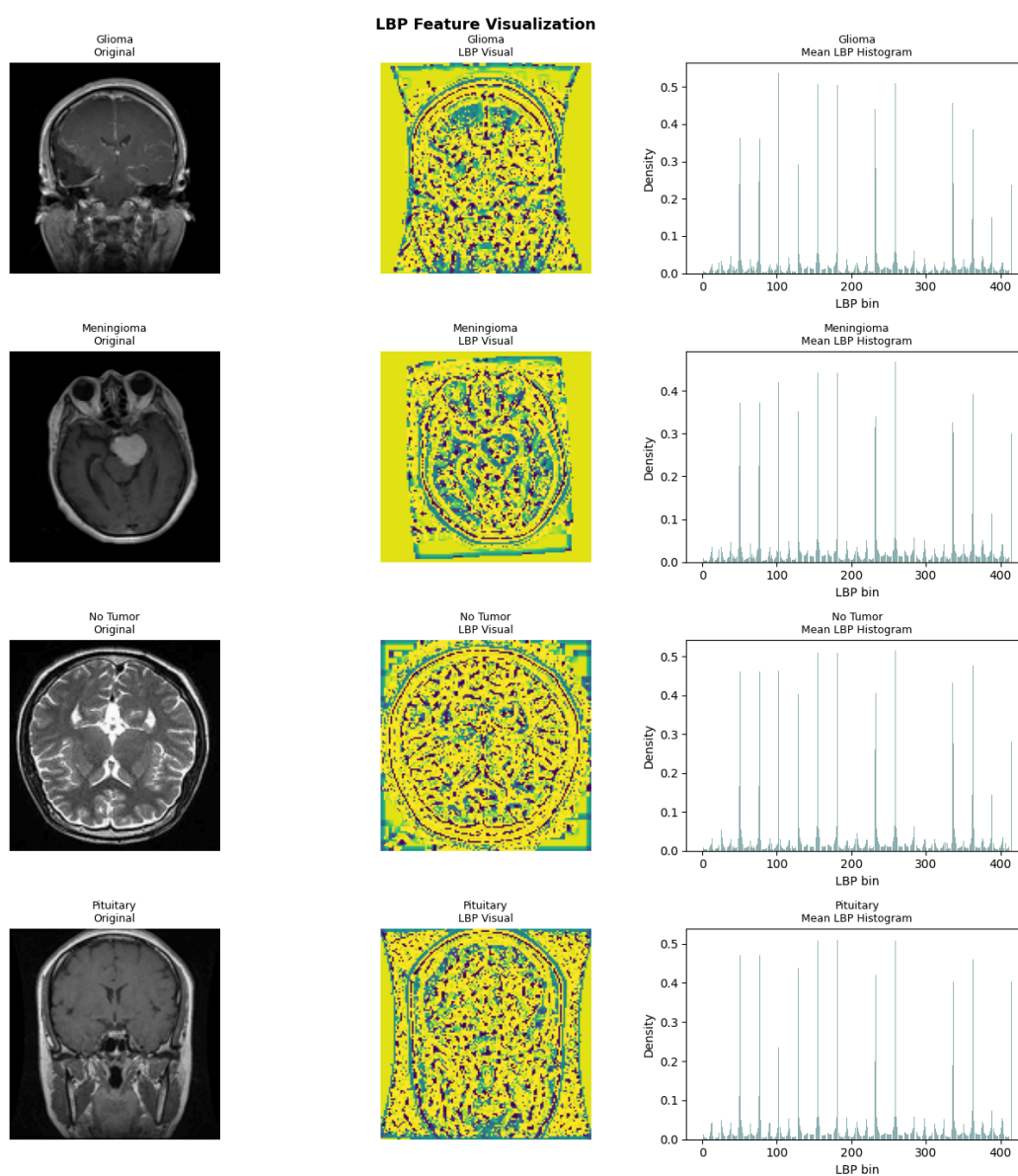
Local Binary Pattern (LBP)

LBP provides complementary information to HOG by capturing local texture information that gradient-based descriptors do not represent well. While HOG summarizes the *direction* of intensity change at each location, LBP summarizes the *pattern* of intensity relationships in each

pixel's neighborhood—comparing each pixel to its circular neighbors and encoding the result as a small integer code. Aggregating these codes into histograms produces a compact texture signature for each region of the image.

This is particularly relevant for our hardest class pair; glioma and meningioma share similar global structure but differ in internal texture – exactly the kind of distinction shape-based features miss but pattern-based descriptors can exploit.

We extract uniform LBP with radius 3 and 24 sampling points, computed independently over a 4×4 spatial grid (26 bins per cell $\times$ 16 cells = 416 dimensions). The spatial grid retains coarse positional information that a single image-wide histogram would discard, allowing the descriptor to register where in the image a particular texture appears.



The LBP visualization shows visibly different texture maps across classes, and the mean per-class LBP histograms differ in the relative heights of dominant bins, confirming that LBP is picking up class-discriminative texture content.

ResNet-18

HOG and LBP rely on hand-designed rules regarding visual information: gradient direction at one scale, local intensity patterns at another. Some important distinctions, however, such as subtle morphological differences between visually similar tumor types are not easily captured in these methods. To address this, we incorporated learned semantic features, using a frozen ResNet-18 pretrained on ImageNet as a fixed feature extractor.

We replace the final classification layer with an identity mapping, which makes the network output the 512-dimensional embedding from its global average pooling layer instead of class scores. ResNet-18 was trained on 3-channel 224×224 RGB images with ImageNet normalization, so we upsample our 128×128 grayscale images to 224×224 , copy the single channel three times to fake an RGB input, and apply the standard ImageNet mean and standard deviation. The same preprocessed images feed all three feature extractors, so the comparison across feature types is on the same underlying data. We extract ResNet features once in GPU batches and reuse them for every downstream experiment.

Unlike the hand-engineered descriptors, these features encode mid- and high-level patterns learned from large-scale natural-image pretraining, providing representations that go beyond simple gradient or texture information from pixel intensities. As a result, ResNet features offer a complementary view of the data that may help capture more complex or subtle distinctions, especially in separating visually similar tumor types.

Final Feature Selection

We chose HOG, LBP, and ResNet-18 because they capture complementary aspects of the images. HOG focuses on global structure and edges, LBP captures local texture, and ResNet-18 provides more complex, learned representations. This aligns with our exploratory analysis, where some classes were separable by overall shape while others required finer texture or higher-level patterns. Evaluating all three features allows us to compare simpler, interpretable features against more expressive, deep features under the same modeling setup, and evaluate how each representation impacts classification performance.

PCA and t-SNE Analysis

Intrinsic Dimensionality

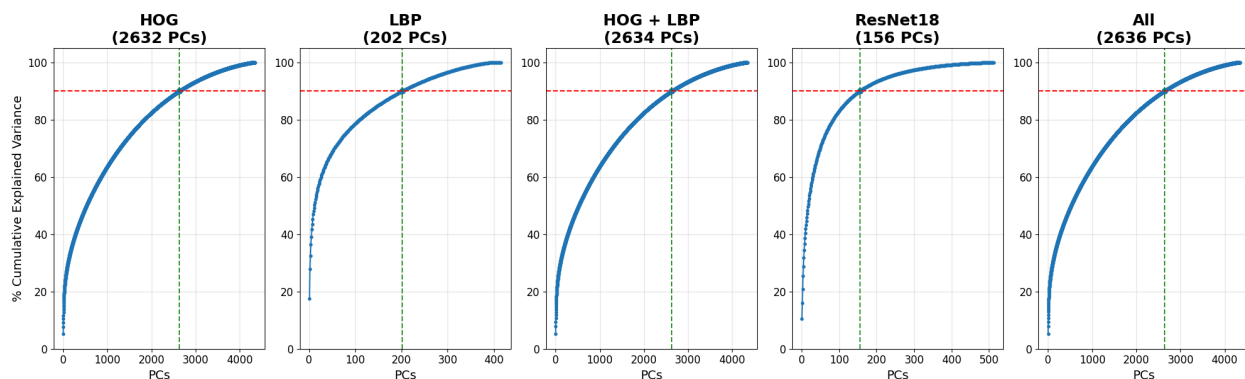
The cumulative explained-variance curves below show how compactly each feature representation can be summarized. ResNet-18 features are by far the most compressible: 156

components capture 90% of the variance, which is roughly 30% of their original 512 dimensions. LBP follows at 202 of 416 components (~49%), while HOG requires 2,632 of its 46,128 components (~6%) to reach the same threshold. The combined feature vectors track HOG closely with 2,634 for HOG+LBP and 2,636 for HOG+LBP+ResNet18 also referred to as AllFeat – unsurprising given that HOG dominates the combined dimensionality.

The shape of each curve is also informative: ResNet-18's curve rises steeply in the first ~50 components before flattening, indicating a small number of high-variance directions carry most of the signal, which is consistent with the embedding structure of a pretrained network. This suggests that the intrinsic dimensionality of the ResNet-18 feature space is relatively low. HOG's curve, by contrast, climbs almost linearly across thousands of components, indicating the variance is distributed across many weakly contributing directions. This demonstrates that HOG has a diffuse representation that is less easily compressed. The combined feature sets closely follow HOG, suggesting that concatenation inherits the inefficiencies of the highest dimensional component rather than producing a more compact embedding.

The differences demonstrated in the PCA analysis helped guide our downstream modeling. On the one hand, diffuse feature spaces benefit from stronger dimensionality reduction to remove non-informative variance. On the other hand, compact representations can retain useful information across more components. This also guided our approach to regularization tuning. Higher dimensional, information rich features often require stronger regularization, whereas reduced feature spaces can be modeled with less. Given this interplay, it is important to tune both PCA dimensions and regularization across the models.

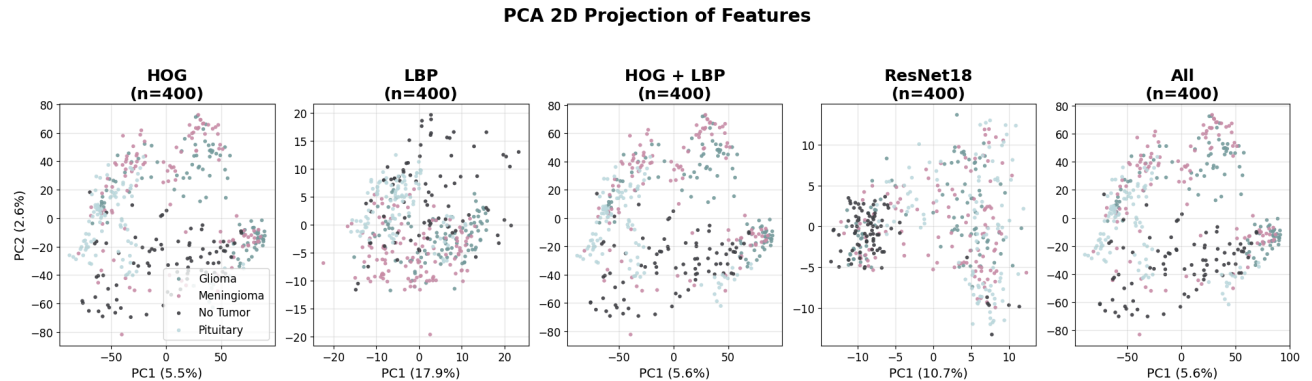
PCA Across Feature Types



Linear Separability

Projecting each feature set onto its top two principal components reveals what class structure is captured by purely linear, variance-maximizing directions. However, PC1 and PC2 explain only a small fraction of total variance for the high-dimensional features (5.5% for HOG, 5.6% for AllFeat), which is too few to expect clean class separation. Therefore, these scatter plots are best read as a qualitative check rather than evidence of true separability. Across all five feature types, classes overlap heavily in 2D, with no representation showing clear linear cluster structure. The most visible pattern is in ResNet-18 (where PC1+PC2 captures 11.7% of

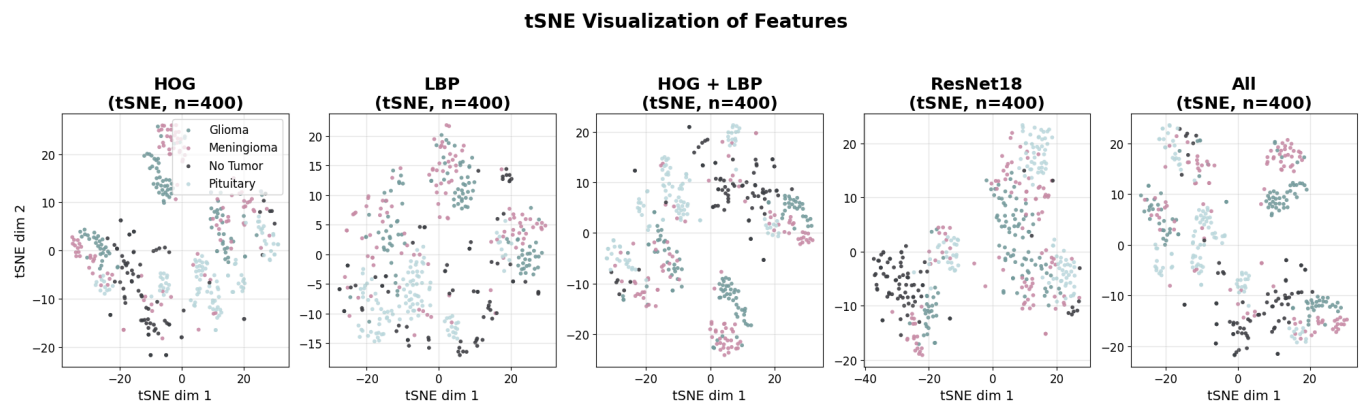
variance): no-tumor points cluster loosely on the left, while glioma, meningioma, and pituitary disperse to the right with substantial overlap. This previews a finding that holds throughout our results: no-tumor is the easiest class to separate from the others, while the three tumor classes are harder to disentangle linearly.



Nonlinear Cluster Structure

Where PCA shows structure along linear, variance maximizing directions, t-SNE shows local cluster structure and can reveal clustering that may exist along nonlinear manifolds. Compared to our PCA projections, t-SNE produces more visible class groupings across all feature types, with same-class points forming small contiguous neighborhoods rather than overlapping clouds. As seen in the plot, however, these clusters are still imperfect with noticeable mixing between classes.

Among the features, ResNet-18 and AllFeat show the most apparent separation, particularly isolating the no-tumor class (black) well from the tumor classes. LBP, in contrast, appears the most diffuse and has substantial overlap among classes. HOG and HOG+LBP show some moderate structure but still have significant mixing among glioma, meningioma, and pituitary.



Together, these observations suggest that meaningful class structure could exist in the feature space but are not cleanly separable via simple linear projections. This motivates two pipeline

design choices: (1) PCA is applied to reduce dimensionality before classification, leveraging the fact that all five feature representations are substantially redundant; and (2) nonlinear classifiers (RBF-kernel SVM in particular) are worth examining to capture the more complex decision boundaries outlined by the t-SNE structure.

Classification Models

Model Selection and Motivation

We use logistic regression and support vector machine (SVM) algorithms to evaluate how different feature representations perform under two standard classification approaches. Logistic regression provides a simple linear baseline, while SVM allows for more flexible, potentially nonlinear decision boundaries.

Logistic Regression

We use multinomial logistic regression as a linear baseline to evaluate how well each feature representation supports class separability. The model learns linear decision boundaries over input features (HOG, LBP, ResNet, and their combinations) and outputs class probabilities via a softmax function.

Due to the high dimensionality of the feature sets (e.g., over 46k dimensions for HOG), we apply PCA to reduce dimensionality before training and improve model stability. Namely, we jointly tune the number of PCA components alongside model hyperparameters to identify the best performing configurations.

Because logistic regression assumes linear separability, its performance is limited by the substantial class overlap as seen in the PCA and t-SNE analysis, especially between glioma and meningioma. Despite this, logistic regression provides a useful and interpretable reference point for comparing the effectiveness of different feature representations.

Support Vector Machine (SVM)

We use an SVM with an RBF kernel to capture nonlinear structure in the feature space, allowing the model to learn complex decision boundaries beyond what linear models can represent. Like in the logistic regression model, we jointly tune the number of PCA components alongside the SVM hyperparameters. We perform a grid search over C and γ to identify the best performing configurations with respect to validation accuracy.

Across feature sets, higher C values and $\gamma = \text{"scale"}$ consistently produced stronger results, which ultimately indicate that more flexible decision boundaries were beneficial. This modeling choice is supported by our earlier analysis; PCA projects show limited linear

separability, while the t-SNE reveals more nonlinear cluster structure. Consequently, SVM with the RBF kernel is well-suited for our tumor classification problem.

Model Evaluation

Evaluation Methodology

We evaluated each combination of feature representation and classifier using a standard three-way data split: a training set for fitting model parameters, a validation set for hyperparameter search, and a held-out test set for final evaluation. The train/validation split was stratified by class to keep per-class proportions consistent, and the test set consisted of the deduplicated Kaggle test partition. To prevent data leakage, both standardization (StandardScaler) and PCA were fit exclusively on training data and then applied unchanged to the validation and test sets.

For each feature configuration, we jointly tuned the number of PCA components and classifier hyperparameters— C and γ for the SVM, and C for logistic regression—ultimately selecting the best model on validation accuracy within each feature configuration. The validation-selected models were then evaluated on the test set to characterize the accuracy-efficiency frontier, with the best test-performing configuration within each feature family highlighted in the per-classifier comparison figures in the accuracy-efficiency section below.

Metrics

We report a complementary set of metrics chosen to expose different failure modes. Test accuracy gives a single headline number for top-line comparison and is appropriate here because the test set is roughly class-balanced. Per-class precision, recall, and F1 are reported via `classification_report` because a top-line accuracy can mask asymmetric errors that matter clinically. Missed glioma and missed pituitary are not interchangeable failures, and per-class metrics make those differences visible. Row-normalized confusion matrices complement the per-class scores by showing which class pairs get confused with which, something not captured by scalar metrics. One-vs-rest ROC curves and per-class AUC, with macro-averaged AUC, characterize each class's separability across all decision thresholds rather than at a single operating point. This is useful because the appropriate threshold for a clinical screening tool would not necessarily be the `argmax` used at prediction time. We also monitor the train-validation gap as a basic overfitting check, particularly for the higher-dimensional feature configurations where the grid search could otherwise reward memorization.

Hyperparameter Search

For each of the five feature configurations (HOG, LBP, ResNet-18, HOG+LBP, AllFeat), we performed a joint grid search over PCA dimensionality and classifier hyperparameters. PCA targets were swept across $\{50, 100, 200, 300, 500, 1000\}$ for HOG and combined feature sets,

{50, 100, 200} for LBP, and {50, 100, 150} for ResNet-18, with ranges chosen to bracket the dimensionalities at which each feature family captures most of its variance. For SVM (RBF kernel), we searched over $C \in \{1, 10, 15, 50\}$ and $\gamma \in \{\text{scale}, 0.01\}$. For logistic regression (multinomial, lbfgs solver, max_iter=1000), we searched over $C \in \{0.0025, 0.005, 0.01, 0.1\}$. The configuration maximizing validation accuracy was selected per (feature family, classifier) pair.

Generalizability

Across our top-performing configurations, validation and test accuracies differed by 2–4 percentage points. We assessed whether this represents genuine generalization, rather than a fortunate split, through several converging signals, all drawn from data already produced by the pipeline.

First, validation and test accuracies track each other closely across the full grid of 48 (feature, classifier, PCA-dimension) configurations, with the broad ranking preserved on both partitions. The search did not surface configurations that succeeded on validation but failed on the held-out test set. Second, PCA dimensionality acted as an implicit regularizer: across feature families, the best-performing dimensions sat at the low-to-mid end of our sweep (100–300 components for HOG and AllFeat), with higher-dimensional configurations consistently underperforming on the test set. This is the pattern expected when capacity-driven overfitting is constrained by the regularization implicit in PCA truncation. Third, the chosen hyperparameters sat on stable plateaus rather than knife-edge optima: at HOG+SVM with PCA=100, for instance, validation accuracy varied by less than half a percentage point across $C \in \{10, 15, 50\}$, indicating performance was not contingent on a brittle setting. Finally, StandardScaler and PCA were both fit only on training data and applied unchanged to validation and test, preventing information from the held-out partitions from leaking into the selection process. MD5-based deduplication independently confirmed no identical images appeared in both training and test partitions.

Several refinements would strengthen this generalization story further. K-fold cross-validation on the training+validation pool would yield a quantitative stability estimate (mean $\pm$ std accuracy across folds) for each headline configuration, complementing the qualitative signals above with a single defensible number per model. The grid search itself could be widened—particularly for SVM γ , where our two-value sweep ($\{\text{'scale'}, 0.01\}$) constrained the search to a narrow region of the hyperparameter space—and could be coupled with a held-out validation fold rather than a single fixed split, reducing dependence on a single fixed partition. Finally, ensembling the top-N configurations rather than reporting a single winner would likely reduce variance from any one model's idiosyncratic fit.

Results

Three key takeaways emerge from our results. First, SVM consistently outperforms logistic regression across all feature representations, indicating that nonlinear decision boundaries are

critical for this task. Second, under SVM, performance across feature sets converges within a narrow range, suggesting that multiple representations capture similar underlying signals in the dataset. Third, combining features yields only marginal gains over individual representations, indicating diminishing returns from feature fusion.

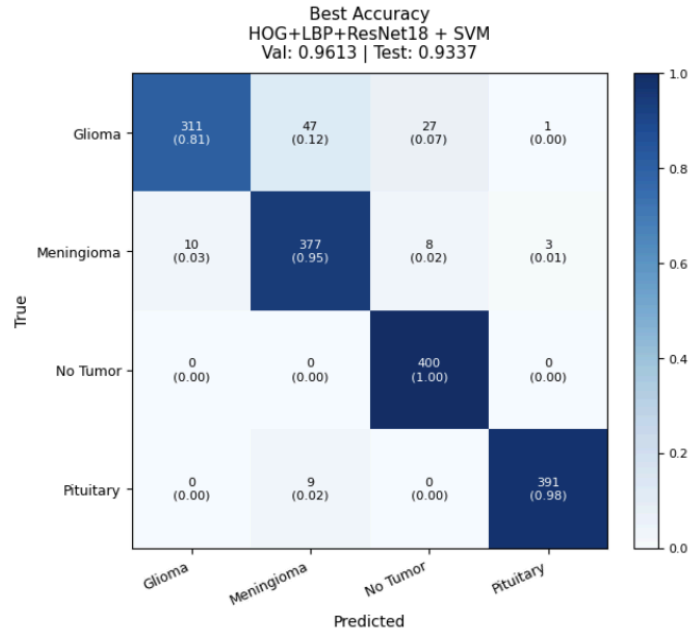
Across the full grid of feature representations, dimensionalities, and classifiers, RBF-kernel SVMs dominated logistic regression on every feature family. The single best-performing configuration was AllFeat features reduced to 100 PCA components and classified with an SVM ($C=15$, $\gamma=\text{'scale'}$), achieving 96.13% validation accuracy and 93.37% test accuracy. The gap between validation and test accuracy is approximately 2.8 percentage points, indicating that hyperparameter selection on the validation set generalized well to the held-out partition. Among logistic regression configurations, the strongest was also an AllFeat configuration but with 500 PCA components at 88.6% test accuracy, roughly five points behind the best SVM result.

The top-performing configurations cluster tightly across all SVM-based feature combinations: AllFeat+SVM (93.4%), HOG+LBP+SVM (93.2%), ResNet-18+SVM (92.9%), and HOG+SVM (92.8%) all fall within one percentage point of each other. This narrow spread suggests that, with a sufficiently expressive classifier, several feature representations capture enough discriminative signal to approach a shared performance ceiling on this dataset.

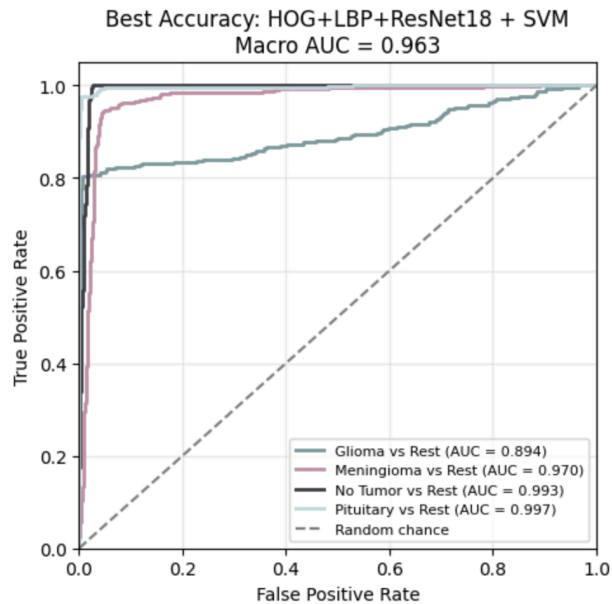
Per-Class Performance

The best-accuracy model showed strongly asymmetric per-class behavior. No-tumor and pituitary classes were nearly fully separable, with no-tumor reaching 100.0% recall (95.8% F1) and pituitary reaching 97.8% recall (98.4% F1). Meningioma achieved 94.7% recall with somewhat lower precision (87.1%), reflecting confusion from other classes being predicted as meningioma. Glioma was the most difficult class, with 96.9% precision but only 80.6% recall (88.0% F1)—the model rarely mistook other classes for glioma, but it missed roughly one in five true glioma scans, most often labeling them as meningioma. This pattern is consistent with our exploratory observation that glioma and meningioma produced nearly identical mean-image signatures, indicating overlapping global structure that requires fine-grained features to disentangle.

The confusion matrix, outputted below, reiterates this per-class asymmetry. The majority of errors occur between glioma and meningioma, while the no-tumor and pituitary classes remain largely isolated along the diagonal. The greatest source of confusion for the model comes from the glioma class, where a substantial portion is predicted as meningioma. Ultimately, this reinforces that the model struggles to distinguish between these two tumor types despite overall high validation and test accuracies. Errors involving no-tumor and pituitary are minimal and further emphasize the separability of the feature space. Overall, the matrix confirms that performance limitations are driven by pairwise confusion rather than a general failure across all classes.



The one-vs-rest ROC curves for our best accuracy model, plotted below, tell a similar story. Per-class AUCs for the best-accuracy model were 0.99 (pituitary), 0.99 (no tumor), 0.97 (meningioma), and 0.89 (glioma), yielding a macro-averaged AUC of 0.96. While pituitary and no tumor are near perfect, the significantly lower glioma curve and AUC demonstrates that the model struggles to consistently separate glioma from other classes across all decision thresholds. This weaker separability and greater overlap in predicted score distributions between glioma and meningioma is consistent with the observed confusion found in the other per-class performance outputs.



Classifier Limitations

The strongest limitation of our final classifier is the persistent confusion between glioma and meningioma. The best-accuracy model misclassifies roughly one in five gliomas as meningiomas, dragging glioma recall to 0.806 despite high precision (0.969). The CNN baseline showed the same pattern (72% glioma recall vs. 99% no-tumor), and the per-class AUC for glioma (0.89) is substantially lower than for the other three classes (≥ 0.97), indicating this is not a thresholding artifact but that the two distributions genuinely overlap in the feature space we extracted. Resolving this likely requires either tumor-localized features (e.g., crops around segmentation masks) or fine-tuned deep features that learn MRI-specific discriminators rather than ImageNet ones.

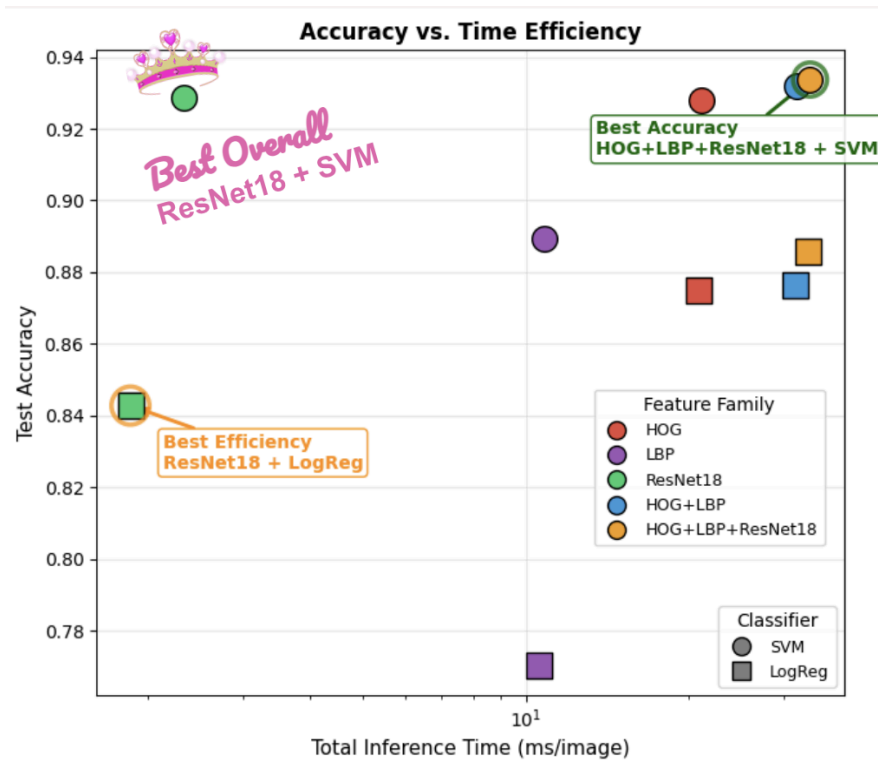
A second limitation is that whole-image classification leaves the model exposed to non-tumor cues. Our exploratory analysis found that the no-tumor class differed systematically from the tumor classes in resolution distribution, source heterogeneity, and the presence of color artifacts; we mitigated the most obvious of these by resizing all images to 128×128 and loading as grayscale, but residual sourcing artifacts (compression, contrast distribution, framing) may still contribute to no-tumor's near-perfect separability. Without segmentation-supervised training, we cannot rule out that some of our 100% no-tumor recall is driven by sourcing rather than anatomy.

Finally, the classifier inherits the limits of its training distribution. The Kaggle dataset is small (~7,000 images), drawn from public sources of unknown provenance, and contains no demographic, scanner, or sequence metadata. We have not validated against an external dataset, so reported test accuracy is best interpreted as in-distribution performance rather than a deployment-ready estimate.

Efficiency-Accuracy Tradeoffs

To analyze efficiency-accuracy tradeoff, we measured per-image inference cost as the sum of two timed components. Feature extraction time was measured by running each extractor on a 100-image subset of the training set and dividing wall-clock time by 100. Classifier inference time was measured by calling predict on the full PCA-reduced test set and dividing by the number of test images. Times for combined feature configurations (HOG+LBP and AllFeat) were computed by summing the individual extractor times rather than re-timing a fused pipeline; this is justified because the extractors are run sequentially with no shared computation. Two methodological caveats apply: timings are single-run rather than averaged across repeated trials, and the PCA transform step is not included in the reported inference cost, since predictions are made on pre-reduced feature matrices. The PCA transform contributes well under a millisecond at our number of feature dimensionalities and does not alter our conclusions below.

Within each (feature family, classifier) pair we selected the PCA dimensionality with the highest test accuracy as the representative model for that pair, yielding ten reference points. Total inference time and test accuracy are plotted on a log-x scatter in the figure below.



Three patterns are visible in the results. First, SVM with the RBF kernel outperforms logistic regression on every feature set, with margins ranging from 4.8 points (AllFeat: 88.6% → 93.4%) to 12 points (LBP: 77.0% → 89.0%). The disproportionately large margin on LBP suggests that the 416-dimensional spatial-LBP histogram is not linearly separable across the four classes and benefits substantially from the RBF kernel's nonlinearity; the smaller gap on the deep features is consistent with ResNet-18 embeddings already lying close to a linearly separable representation.

Feature	Classifier	Accuracy	Speed (ms)
HOG	LR	87.5%	20.881
HOG	SVM	92.8%	21.082
LBP	LR	77%	10.562
LBP	SVM	89%	10.822
ResNet-18	LR	84.3%	1.861
ResNet-18	SVM	92.9%	2.341
HOG + LBP	LR	87.6	31.442
HOG + LBP	SVM	93.2%	31.638
AllFeat	LR	88.6%	33.302
AllFeat	SVM	93.4%	33.495

Second, HOG dominates the inference cost budget. HOG extraction alone accounts for roughly 21 ms per image, while LBP adds approximately 11 ms and ResNet-18 features complete in under 2.5 ms on our hardware. Any configuration containing HOG therefore inherits its latency, which is why HOG+LBP and AllFeat sit on the right side of the scatter regardless of classifier choice. Third, feature concatenation yields diminishing accuracy returns. AllFeat + SVM achieves the top accuracy at 93.4%, but this is only 0.2 points above HOG+LBP + SVM (93.2%) and 0.5 points above ResNet-18+SVM (92.9%), at roughly 14× the inference cost of ResNet-18+SVM.

The practical implication is that ResNet-18 + SVM, highlighted in green in the table above and crowned in the figure, is the optimal operating point for deployment. At 2.34 ms per image it runs nearly an order of magnitude faster than any HOG-based pipeline, while sacrificing only 0.5 percentage points in accuracy relative to the best configuration. The 512-dimensional ResNet-18 embedding appears to capture sufficient discriminative structure that the additional hand-crafted descriptors contribute marginal new information at substantial computational cost. For applications prioritizing maximum accuracy over latency, AllFeat + SVM remains the recommended choice, particularly given accuracy's importance in a tumor classification context. For interactive or real-time screening contexts, ResNet-18 + SVM offers the most favorable tradeoff. Logistic regression is not competitive at any operating point: its best result (AllFeat + LR, 88.6%) is surpassed by SVM on every feature set, including LBP alone.

Future Improvements

Several directions could meaningfully extend this work. The most direct improvement would likely be fine-tuning ResNet on the MRI dataset rather than using it as a frozen ImageNet feature extractor; this would let the network learn representations specifically tuned to the textures and structures relevant to brain tumor classification rather than relying on features optimized for natural images.

A second priority is incorporating segmentation supervision to isolate tumor regions before classification. Our heuristic localization approaches all failed in the absence of pixel-level annotations, but a dataset with segmentation masks, or a pretrained MRI segmentation model, would let us crop to the region of interest and remove background and skull-related signal that the current pipeline must work around.

Beyond architecture changes, scaling the dataset is likely the single highest-leverage improvement: 7,000 images is a small sample for a four-way medical classification task, and additional data drawn from diverse scanners and patient populations would directly improve generalization. Relatedly, our feature fusion strategy could be more sophisticated. Naive concatenation lets HOG dominate the combined representation by sheer dimensionality (46,128 dimensions vs. 416 and 512 for LBP and ResNet), which limits the contribution of the complementary descriptors. Per-feature normalization or learned fusion weights would let each representation contribute more equitably.

Finally, evaluation could be made more clinically meaningful in two ways. Reporting metrics aligned with diagnostic priorities—particularly per-class recall, since missed tumors carry far higher cost than false positives—would more accurately reflect deployment value than headline accuracy. And benchmarking against expert human readers, such as radiologists or trained clinicians, would contextualize whether our 93% accuracy is competitive with human performance or substantially below it, which is ultimately the question that determines whether classical computer vision is a viable approach for this problem.

Conclusion

In this work, we evaluated how different feature representations—HOG, LBP, and pretrained ResNet-18—impact brain tumor classification performance under a consistent modeling framework. Across experiments, nonlinear classifiers consistently outperformed linear models, and multiple feature representations achieved similar levels of performance. While combining features yielded the highest accuracy, the gains were marginal relative to the additional computational cost.

More broadly, our findings suggest that pretrained deep representations can capture much of the discriminative signal in this dataset, with additional handcrafted features providing diminishing returns. This highlights the importance of balancing model complexity with efficiency, rather than assuming that more complex feature pipelines will yield meaningful improvements.

Appendix

Supplementary Experiments

ResNet-18 vs ResNet-50

To inform the choice of pretrained backbone for our deep feature extractor, we ran a separate comparison of ResNet-18 and ResNet-50 as frozen feature extractors before settling on the final pipeline. Both backbones used ImageNet-pretrained weights with the final classification head replaced by an identity mapping, yielding 512-dimensional features for ResNet-18 and 2,048-dimensional features for ResNet-50. Each feature set was evaluated with two downstream classifiers—an RBF-kernel SVM and logistic regression—on a stratified 80/20 train/validation split of the Kaggle training set (4,480/1,120 images) and the original Kaggle test partition (1,600 images, 400 per class).

This experiment used an earlier version of our preprocessing pipeline that resized images to 224×224 and applied CLAHE contrast normalization, without the MD5-based deduplication step later adopted in the final pipeline. The comparison is therefore not directly numerically comparable to the headline results in the Features section of this report, but the relative

comparison between backbones is informative: ResNet-18 + RBF SVM achieved 95.36% validation and 91.69% test accuracy, outperforming ResNet-50 + RBF SVM (93.66% / 90.06%) despite having a quarter as many feature dimensions and roughly an eighth of the SVM fit time (1.07s vs 8.15s). Per-class results showed the same difficulty pattern observed throughout our work: pituitary and no-tumor were easiest (0.97–1.00 recall), meningioma intermediate (0.93), and glioma hardest (0.76 recall).

We interpret the stronger ResNet-18 performance as evidence that the smaller backbone's lower-dimensional feature space is better matched to a dataset of this size; ResNet-50's higher-capacity representations likely over-parameterize the downstream classifier without gaining additional discriminative signal. On this basis, ResNet-18 was selected as the deep feature extractor for the final pipeline.

Image Flipping

To assess robustness to spatial transformations, we evaluated the trained model on a horizontally flipped version of the test set, while preserving the original training and validation data. This experiment isolates the effect of left–right orientation on model predictions without introducing data leakage.

Model performance remained largely consistent under this transformation, suggesting that the classifier has learned features that are relatively invariant to horizontal reflection. Given the approximate bilateral symmetry of brain structures, this result is consistent with expectations and provides additional evidence that the model is capturing semantically meaningful patterns rather than overfitting to positional cues.

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Note: Generative AI tools (e.g., Claude) were used for outlining, ideation, and phrasing, but all analysis, modeling decisions, and interpretations were conducted independently.